InstaGAN: Instance-aware Image-to-Image Translation

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Problem: Image-to-image translation

Problem of interest:

- Transform target scene elements while preserving the original content
- Prior methods often fail on challenging tasks, e.g., when the translation involves large changes in shape, or the image contains multiple instances

Our goal:

- Extend image-to-image translation toward multi-instance transfiguration
- Many possible applications, e.g., virtual try-on of fashion items



(a) jeans→skirt (b) sheep→giraffe

Main idea: Instance-aware GAN (InstaGAN)

Main idea:

- Incorporate the additional instance attributes (e.g., segmentation masks[a])
- Since an image can contain multiple instances, we should learn a joint mapping of an image x and the set of instance attributes $\boldsymbol{a} = \{a_i\}_{i=1}^N$, i.e., learn a mapping between attribute-augmented spaces

$$(x, \boldsymbol{a}) \in \mathcal{X} \times \mathcal{A} \rightleftharpoons (y, \boldsymbol{b}) \in \mathcal{Y} \times \mathcal{B}$$

Our contribution:

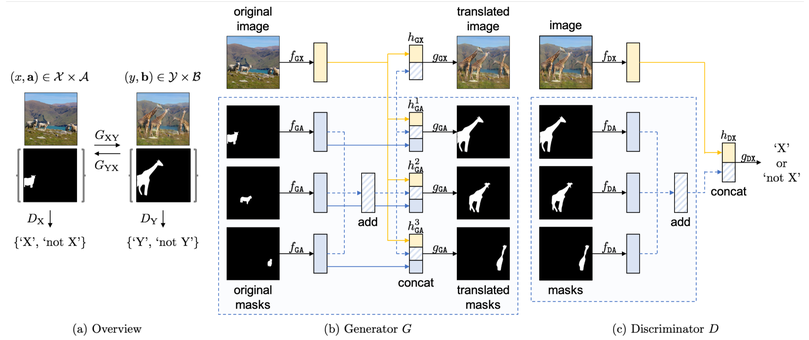
- We propose 1) neural architecture, 2) loss function, and 3) translation scheme to learn a joint mapping of an image and the set of instances

[a] Segmentation mask may be a good representation for instance shapes, as it contains object boundaries while ignoring other details such as color

InstaGAN architecture & loss function

InstaGAN architecture:

- Jointly train two generators G_{XY}, G_{YX}, and two discriminators D_{X}, D_{Y}
- Each network encodes both an image and the set of instance masks
- Here, the order of instance masks should not affect the translated results, i.e., image is *permutation invariant*, and mask is *permutation equivariant*
- We sum & concat the features of all set elements to achieve the property



(a) Overview (b) Generator G (c) Discriminator D

Loss function:

- Image-to-image translation = domain loss + content loss[a]
- We propose new content loss utilizing the instance masks[b]

$$\mathcal{L}_{\texttt{ctx}} = \|w(\boldsymbol{a}, \boldsymbol{b}') \odot (x - y')\|_1] + \|w(\boldsymbol{b}, \boldsymbol{a}') \odot (y - x')\|_1$$

- Total loss of InstaGAN is

$$\mathcal{L}_{\texttt{InstaGAN}} = \underbrace{\mathcal{L}_{\texttt{LSGAN}}}_{\texttt{GAN (domain) loss}} + \underbrace{\lambda_{\texttt{cyc}}\mathcal{L}_{\texttt{cyc}} + \lambda_{\texttt{idt}}\mathcal{L}_{\texttt{idt}} + \lambda_{\texttt{ctx}}\mathcal{L}_{\texttt{ctx}}}_{\texttt{content loss}}$$

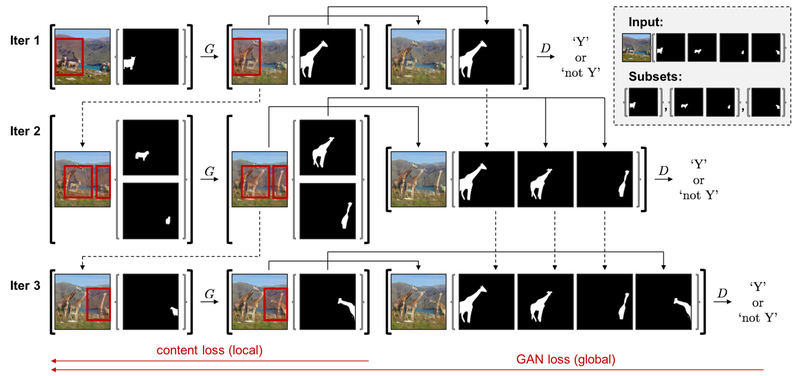
[a] We use GAN loss for domain loss, and cycle-consistency loss for content loss
[b] Enforce to translate instances only, while keeping outside of them (i.e., background)

Sequential mini-batch translation

Motivation: GPU memory linearly increases with the number of instances

Sequential inference scheme:

- Divide the set of instances into mini-batches, and sequentially infer them
- Final output aggregates all the translated masks to align with the image



Sequential training technique:

- Apply content loss to the intermediate samples of current mini-batch, and apply GAN loss to the samples of aggregated mini-batches[a]
- Sequential version of the training loss of InstaGAN is

$$\mathcal{L}_{\texttt{InstaGAN-SM}} = \sum_{m=1}^{M} \mathcal{L}_{\texttt{LSGAN}}((x, \boldsymbol{a}), (y'_m, \boldsymbol{b}'_{1:m})) + \mathcal{L}_{\texttt{content}}((x_m, \boldsymbol{a}_m), (y'_m, \boldsymbol{b}'_m))$$

Effects of the sequential mini-batch translation:

- Improve the many-instance results by training arbitrary many instances
- Improve the few-instance results due to its data augmentation effect

[a] Detach every m-th iteration of training, so that only a fixed GPU memory is required

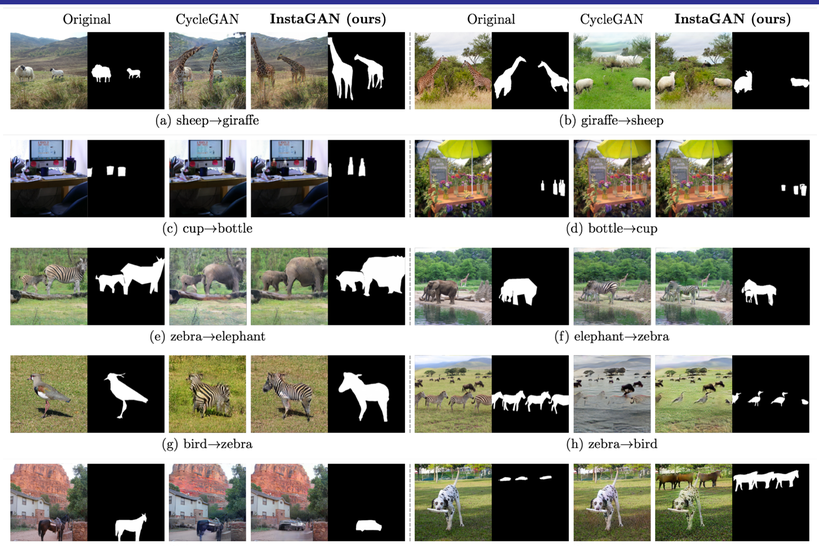
Experimental results: Multi-instance transfiguration

Image-to-image translation results:

- Our method shows impressive results on many multi-instance transfiguration tasks, sampled from fashion (CCP, MHP) and COCO datasets



(a) jeans→skirt (b) skirt→jeans (c) shorts→pants (d) pants→shorts (e) pants→skirt (f) skirt→pants



(a) sheep→giraffe (b) giraffe→sheep (c) cup→bottle (d) bottle→cup (e) zebra→elephant (f) elephant→zebra (g) bird→zebra (h) zebra→bird (i) horse→car (j) car→horse

- Our method can control the instances to translate by conditioning input



- Our method generalizes to unseen samples by using predicted masks



- Our method produces temporally coherent results on video translation



Original Translated